

AWS AI Practitioner

Practice Questions & Answers

73

Total Questions

6

Exam Domains

100%

Answers Explained

AI/ML Concepts	3 questions (4%)
AWS Infrastructure	1 questions (1%)
AWS Security	6 questions (8%)
AWS Services	20 questions (27%)
Generative AI	18 questions (24%)
ML Concepts	14 questions (19%)
Responsible AI	11 questions (15%)

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Q1

AI/ML CONCEPTS

A company built a deep learning model for object detection and deployed the model to production. Which AI process occurs when the model analyzes a new image to identify objects?

- A Bias correction
- B Model deployment
- C Training

D Inference**CORRECT ANSWER****Option D — Inference****EXPLANATION**

When a deployed model analyzes new, unseen data (like a new image) to produce predictions or identify objects, that process is called Inference. The model is applying what it learned during training to generate an output — no learning or weight updates are happening. Training is what happened earlier. Model deployment is the act of making the model available. Bias correction is a technique applied during training or preprocessing.

Q2

RESPONSIBLE AI

A company makes forecasts each quarter to decide how to optimize operations to meet expected demand. The company uses ML models to make these forecasts. An AI practitioner is writing a report about the trained ML models to provide transparency and explainability to company stakeholders. What should the AI practitioner include in the report to meet the transparency and explainability requirements?

A Partial dependence plots (PDPs)

B Sample data for training

C Code for model training

D Model convergence tables

CORRECT ANSWER

Option A — Partial dependence plots (PDPs)

EXPLANATION

PDPs show how each input feature influences the model's predictions, making it easy for stakeholders to understand why the model forecasts what it does — in a visual, interpretable way. This directly addresses transparency and explainability requirements for non-technical audiences. Sample training data doesn't explain how the model reasons. Code is opaque to business stakeholders. Model convergence tables are technical metrics about the training process, not about explainability.

Q3

AWS SERVICES

A company manually reviews all submitted resumes in PDF format. As the company grows, the company expects the volume of resumes to exceed the company's review capacity. The company needs an automated system to convert the PDF resumes into plain text format for additional processing. Which AWS service meets this requirement?

- A Amazon Lex
- B Amazon Transcribe
- C Amazon Textract**
- D Amazon Personalize

CORRECT ANSWER**Option C — Amazon Textract****EXPLANATION**

Amazon Textract is specifically designed to extract text and data from scanned documents and PDFs, making it the perfect fit for converting PDF resumes into plain text. Amazon Lex builds conversational chatbots. Amazon Transcribe converts audio/speech to text, not PDFs. Amazon Personalize is a recommendation engine.

Q4

AWS
INFRASTRUCTURE

A company needs to build its own large language model (LLM) based on only the company's private data. The company is concerned about the environmental effect of the training process. Which Amazon EC2 instance type has the **LEAST** environmental effect when training LLMs?

A Amazon EC2 P series

B Amazon EC2 C series

C Amazon EC2 G series

D Amazon EC2 Trn series

CORRECT ANSWER

Option D — Amazon EC2 Trn series

EXPLANATION

Amazon EC2 Trn (Trainium) instances are purpose-built by AWS specifically for training deep learning and LLMs. Because they are custom-designed for this exact workload, they achieve significantly better energy efficiency per training job compared to general-purpose GPU instances, meaning less power consumption and a lower carbon footprint. P series instances are general-purpose GPU instances. C series compute-optimized CPU instances would require far more time/energy. G series instances are optimized for graphics and inference, not training.

Q5

GENERATIVE AI

A company needs to choose a model from Amazon Bedrock to use internally. The company must identify a model that generates responses in a style that the company's employees prefer. What should the company do to meet these requirements?

- A Use public model leaderboards to identify the model
- B Evaluate the models by using a human workforce and custom prompt datasets**
- C Evaluate the models by using built-in prompt datasets
- D Use the model InvocationLatency runtime metrics in Amazon CloudWatch when trying models

CORRECT ANSWER

Option B — Evaluate the models by using a human workforce and custom prompt datasets

EXPLANATION

Since the requirement is about generating responses in a style that employees prefer, this is a subjective, human judgment question. Using a human workforce with custom prompts relevant to the company's actual use cases is the only way to accurately measure style preference. Public leaderboards measure general benchmarks, not company-specific style preferences. Built-in prompt datasets are generic and won't reflect the company's specific domain. CloudWatch latency metrics measure response speed, not style preference.

Q6

GENERATIVE AI

A company uses a foundation model (FM) from Amazon Bedrock to develop a chat interface for the company's product manuals. The company wants to fine-tune the model to be more accurate by using the company's data. Which strategy will successfully fine-tune the model?

- A Train the model on journals and textbooks
- B Prepare the training dataset by creating a .txt file that contains multiple lines in .csv format
- C Provide labeled data with the prompt field and the completion field**
- D Purchase Provisioned Throughput for Amazon Bedrock

CORRECT ANSWER**Option C — Provide labeled data with the prompt field and the completion field****EXPLANATION**

Fine-tuning a foundation model in Amazon Bedrock requires a correctly formatted training dataset in JSONL format where each record contains a prompt field (the input) and a completion field (the expected output). Training on journals and textbooks describes pre-training, not fine-tuning on company-specific data. Amazon Bedrock fine-tuning does not accept .txt or .csv files. Provisioned Throughput is about reserving inference capacity, not model training.

Q7

AWS SERVICES

A company has large input data sizes up to 1 GB and processing times up to 1 hour. The company needs near real-time latency. Which SageMaker inference option meets these requirements?

- A Batch transform
- B Real-time inference
- C Asynchronous inference**
- D Serverless inference

CORRECT ANSWER**Option C — Asynchronous inference****EXPLANATION**

Amazon SageMaker Asynchronous inference is specifically designed for large input payloads (up to 1 GB), long processing times (up to 1 hour), and near real-time results. Batch Transform processes large datasets offline with no latency expectation. Real-time inference has a 6 MB payload limit and will time out on long processing jobs. Serverless inference is suited for small payloads and short processing times.

Q8

GENERATIVE AI

A company wants to build a generative AI application by using Amazon Bedrock and needs to choose a foundation model (FM). The company wants to know how much information can fit into one prompt. Which consideration will inform the company's decision?

A Model size

B Context window

C Temperature

D Batch size

CORRECT ANSWER**Option B — Context window****EXPLANATION**

The context window defines the maximum amount of text (measured in tokens) that a model can process in a single prompt, including both the input and the generated output. Model size refers to the number of parameters and does not directly define how much text fits in a prompt. Temperature controls output randomness. Batch size refers to how many requests are processed together during training or batch inference.

Q9

RESPONSIBLE AI

A company wants to build an interactive application for children that generates new stories based on classic stories. The company wants to use Amazon Bedrock and needs to ensure that the results and topics are appropriate for children. Which AWS service or feature will meet these requirements?

- A Amazon Rekognition
- B Guardrails for Amazon Bedrock**
- C Amazon Bedrock playgrounds
- D Agents for Amazon Bedrock

CORRECT ANSWER**Option B — Guardrails for Amazon Bedrock****EXPLANATION**

Guardrails for Amazon Bedrock allows companies to define content filtering policies that block harmful, inappropriate, or off-topic content from both inputs and outputs. For a children's application, guardrails can filter out violence, adult content, hate speech, and topics inappropriate for children. Amazon Rekognition analyzes images and videos, not text generated by an LLM. Bedrock Playgrounds is a testing interface, not a production feature for content enforcement. Agents are for multi-step autonomous workflows, not content safety filtering.

Q1
0

AWS SERVICES

A company wants to build an ML model by using Amazon SageMaker. The company needs to share and manage variables for model development across multiple teams. Which SageMaker feature meets these requirements?

A Amazon SageMaker Data Wrangler

B Amazon SageMaker Model Cards

C Amazon SageMaker Clarify

D Amazon SageMaker Feature Store

CORRECT ANSWER

Option D — Amazon SageMaker Feature Store

EXPLANATION

Amazon SageMaker Feature Store is a centralized repository specifically designed to store, share, and manage ML features (variables) across multiple teams and models. Data Wrangler is a data preparation tool for cleaning and processing raw data, not for sharing features. Model Cards are used to document model information for transparency. Clarify is used to detect bias and provide explainability reports.

Q1
1

ML CONCEPTS

A company wants to classify human genes into 20 categories based on gene characteristics. The company needs an ML algorithm to document how the inner mechanism of the model affects the output. Which ML algorithm meets these requirements?

A Neural networks

B Decision trees

C Logistic regression

D Linear regression

CORRECT ANSWER**Option B — Decision trees****EXPLANATION**

A decision tree handles multi-class classification (20 categories) natively and is inherently explainable and transparent. Every decision the model makes can be traced through a clear sequence of if-then rules, making it easy to document exactly how input gene characteristics lead to each classification outcome. Neural networks are black boxes that are extremely difficult to interpret. Logistic regression is a binary classifier by default and less suited for documenting inner decision logic across 20 categories. Linear regression predicts continuous numerical values, not categories.

**Q1
2****AWS SECURITY**

A company uses a foundation model (FM) on Amazon Bedrock. The FM needs to access encrypted data stored in an Amazon S3 bucket encrypted with SSE-S3. The FM encounters a failure when attempting to access the S3 bucket data. Which solution will meet these requirements?

- A** Ensure that the S3 data does not contain sensitive information
- B** Ensure that the role that Amazon Bedrock assumes has permission to decrypt data with the correct encryption key
- C** Use prompt engineering techniques to tell the model to look for information in Amazon S3
- D** Set the access permissions for the S3 buckets to allow public access to enable access over the internet

CORRECT ANSWER

Option B — Ensure that the role that Amazon Bedrock assumes has permission to decrypt data with the correct encryption key

EXPLANATION

When Amazon Bedrock accesses an S3 bucket, it does so by assuming an IAM role. If that role lacks the necessary S3 and decryption permissions, access will fail. The IAM role needs s3:GetObject permission on the bucket. Removing sensitive data doesn't fix a permissions failure. Prompt engineering influences model responses, not authentication or data access. Making encrypted private data publicly accessible is a serious security anti-pattern.

Q1
3

GENERATIVE AI

A company wants to create an application by using Amazon Bedrock. The company has a limited budget and prefers flexibility without long-term commitment. Which Amazon Bedrock pricing model meets these requirements?

- A Model customization
- B Provisioned Throughput
- C Spot Instance

D On-Demand

CORRECT ANSWER

Option D — On-Demand

EXPLANATION

Amazon Bedrock's On-Demand pricing model charges based on actual usage (per token/per API call) with no upfront commitment and no long-term contract, making it perfect for a company with a limited budget that wants flexibility. Model customization refers to fine-tuning capability, not a pricing model. Provisioned Throughput requires purchasing a committed throughput level for a fixed term. Spot Instances is an Amazon EC2 pricing concept, not an Amazon Bedrock pricing model.

Q1
4**AWS SECURITY**

A company wants to develop a large language model (LLM) application by using Amazon Bedrock and customer data stored in Amazon S3. The company's security policy states that each team can access data for only the team's own customers. Which solution will meet these requirements?

- A** Create one Amazon Bedrock role with full Amazon S3 access. Create IAM roles for each team with access to only each team's customer folders
- B** Redact personal data in Amazon S3. Update the S3 bucket policy to allow team access to customer data
- C** Create a custom service role with Amazon S3 access. Ask teams to specify the customer name on each Amazon Bedrock request
- D** Create an Amazon Bedrock custom service role for each team that has access to only the team's customer data

CORRECT ANSWER

Option D — Create an Amazon Bedrock custom service role for each team that has access to only the team's customer data

EXPLANATION

Creating a per-team Bedrock service role scoped to only that team's S3 data enforces least privilege at the IAM role level. Even if a team member makes an incorrect request, the IAM boundary prevents access to another team's data. Option A creates one Bedrock role with full S3 access, violating least privilege at the service level. Redacting personal data doesn't enforce access segmentation — teams could still read each other's data. Asking teams to self-report the customer name is not a security control; it relies on human honesty with no technical enforcement.

Q1
5

ML CONCEPTS

A company wants to develop an educational game where users answer questions such as: 'A jar contains six red, four green, and three yellow marbles. What is the probability of choosing a green marble from the jar?' Which solution meets these requirements with the LEAST operational overhead?

- A Use unsupervised learning to create a model that will estimate probability density
- B Use supervised learning to create a regression model that will predict probability
- C Use reinforcement learning to train a model to return the probability
- D Use code that will calculate probability by using simple rules and computations**

CORRECT ANSWER

Option D — Use code that will calculate probability by using simple rules and computations

EXPLANATION

Probability problems follow deterministic mathematical rules — the answer is always exact and computable with a simple formula ($4 \text{ out of } 13 = 4/13$). Writing code requires no training data, no model, no infrastructure, and no ML expertise. Unsupervised learning finds hidden patterns in unlabeled data, not rule-based calculations. A supervised regression model would require a large labeled training dataset, model training, and deployment — massive overhead for a problem with an exact mathematical solution. Reinforcement learning is designed for sequential decision-making, not simple probability calculation.

Q1
6

GENERATIVE AI

A company wants to make a chatbot to help customers solve technical problems without human intervention. The chatbot needs to produce responses that adhere to company tone. Which solution meets these requirements?

- A Define a higher number for the temperature parameter
- B Use batch inferencing to process detailed responses
- C Set a low limit on the number of tokens the FM can produce

D Experiment and refine the prompt until the FM produces the desired responses

CORRECT ANSWER

Option D — Experiment and refine the prompt until the FM produces the desired responses

EXPLANATION

Prompt engineering is the most direct and practical way to guide a foundation model to produce responses that match a specific tone, style, and purpose. By iteratively crafting and refining the system prompt, the model's output can be shaped without any retraining. Increasing temperature makes responses more random and creative — the opposite of consistent company tone. Batch inferencing processes multiple requests offline and has nothing to do with response tone. Restricting token count would truncate responses, potentially cutting off answers mid-sentence.

Q1
7

GENERATIVE AI

A company uses a large language model (LLM) on Amazon Bedrock for sentiment analysis. The company needs the LLM to produce more consistent responses to the same input prompt. Which adjustment to an inference parameter should the company make?

A Increase the maximum generation length

B Decrease the temperature value

C Decrease the length of output tokens

D Increase the temperature value

CORRECT ANSWER

Option B — Decrease the temperature value

EXPLANATION

Temperature controls the randomness of a model's output. A lower temperature (closer to 0) makes the model more deterministic, meaning it will produce more consistent, predictable responses to the same input prompt every time. For sentiment analysis, the same text should reliably produce the same sentiment classification. Increasing maximum generation length controls how long the response can be, not consistency. Decreasing output tokens shortens the response but has no effect on consistency. Increasing the temperature does the opposite — it increases randomness and creates more varied, inconsistent outputs.

Q1
8

GENERATIVE AI

A company wants to use a pre-trained generative AI model to generate content for its marketing campaigns. The company needs to ensure that the generated content aligns with the company's brand voice and messaging requirements. Which solution meets these requirements?

- A** Increase the model's complexity by adding more layers to the model's architecture
- B** Select a large, diverse dataset to pre-train a new generative model
- C** Create effective prompts that provide clear instructions and context to guide the model's generation
- D** Optimize the model's architecture and hyperparameters to improve the model's overall performance

CORRECT ANSWER

Option C — Create effective prompts that provide clear instructions and context to guide the model's generation

EXPLANATION

Prompt engineering is the most direct and cost-effective solution since the company is using a pre-trained model. A well-crafted prompt can specify the brand voice, tone, target audience, and messaging guidelines. Adding layers to a pre-trained model's architecture is not feasible post-training. Pre-training a new generative model is enormously expensive and contradicts the premise of using a pre-trained model. Hyperparameter optimization happens during training, not at inference time with a pre-trained model.

Q1
9**AWS SECURITY**

A company wants to use AI to protect its application from threats. The AI solution needs to check if an IP address is from a suspicious source. Which solution meets these requirements?

- A** Create a fraud forecasting system
- B** Develop an anomaly detection system
- C** Create a natural language processing (NLP) named entity recognition system
- D** Build a speech recognition system

CORRECT ANSWER**Option B — Develop an anomaly detection system****EXPLANATION**

An anomaly detection system identifies patterns that deviate significantly from normal or expected behavior. Checking whether an IP address comes from a suspicious source — such as a known malicious range, unusual geographic location, or bot-associated data center — is a classic anomaly detection use case. A fraud forecasting system predicts the likelihood of fraudulent financial transactions and is not designed to evaluate IP reputation. NLP named entity recognition extracts entities like names and locations from text — it has no relevance to IP address analysis. Speech recognition converts audio to text and is entirely unrelated to IP-based threat detection.

Q2
0

AWS SERVICES

A company wants to use language models to create an application for inference on edge devices. The inference must have the lowest latency possible. Which solution will meet these requirements?

- A Deploy optimized large language models (LLMs) on edge devices
- B Incorporate a centralized small language model (SLM) API for asynchronous communication with edge devices
- C Incorporate a centralized large language model (LLM) API for asynchronous communication with edge devices
- D Deploy optimized small language models (SLMs) on edge devices**

CORRECT ANSWER

Option D — Deploy optimized small language models (SLMs) on edge devices

EXPLANATION

Deploying optimized SLMs on-device eliminates network round-trips entirely (inference happens locally) and uses small, efficient models suited for edge hardware. LLMs deployed on edge devices are too large and computationally intensive for most edge hardware. A centralized SLM API, even though the model is smaller, introduces network latency, queuing delays, and connectivity dependency. A centralized LLM API has both large model overhead and network round-trip latency — the worst of both worlds.

Q2
1

GENERATIVE AI

A company wants to use large language models (LLMs) with Amazon Bedrock to develop a chat interface for the company's product manuals stored as PDF files. Which solution meets these requirements **MOST** cost-effectively?

A

Upload PDF documents to an Amazon Bedrock knowledge base and use the knowledge base to provide context when users submit prompts

B

Use prompt engineering to add all the PDF files as context to the user prompt when the prompt is submitted

C

Use prompt engineering to add one PDF file as context to the user prompt when the prompt is submitted

D

Use all the PDF documents to fine-tune a model with Amazon Bedrock

CORRECT ANSWER

Option A — Upload PDF documents to an Amazon Bedrock knowledge base and use the knowledge base to provide context when users submit prompts

EXPLANATION

Amazon Bedrock Knowledge Bases implement RAG — only the most relevant chunks of the manuals are retrieved and passed as context for each query. This means the LLM processes a small, relevant portion per request rather than entire documents, making it highly cost-effective. Adding all PDF files as context in every prompt consumes enormous tokens per request. Adding one PDF per prompt still injects an entire document regardless of relevance. Fine-tuning is expensive (training costs + Provisioned Throughput requirement) and not the right tool for document Q&A.;

Q2
2

AWS SERVICES

A digital devices company wants to predict customer demand for memory hardware. The company does not have coding experience or knowledge of ML algorithms. Which solution will meet these requirements?

- A** Import the data into Amazon SageMaker Data Wrangler. Create ML models using SageMaker built-in algorithms
- B** Store the data in Amazon S3. Create ML models using Amazon SageMaker built-in algorithms
- C** Import the data into Amazon SageMaker Data Wrangler. Build ML models using an Amazon Personalize Trending-Now recipe
- D** Import the data into Amazon SageMaker Canvas. Build ML models by selecting the values in the data from SageMaker Canvas

CORRECT ANSWER

Option D — Import the data into Amazon SageMaker Canvas. Build ML models by selecting the values in the data from SageMaker Canvas

EXPLANATION

Amazon SageMaker Canvas is specifically designed for users without coding experience or ML knowledge. It provides a no-code, point-and-click visual interface where business users can import data, select target columns, and build ML models including time-series demand forecasting without writing any code. Data Wrangler + built-in algorithms still requires technical knowledge to configure ML training jobs. S3 + built-in algorithms requires coding knowledge. Amazon Personalize is a recommendation engine, not a demand forecasting tool.

Q2
3**AWS SECURITY**

A financial institution is using Amazon Bedrock to develop an AI application. The application is hosted in a VPC. To meet regulatory compliance standards, the VPC is not allowed access to any internet traffic. Which AWS service or feature will meet these requirements?

- A** Amazon Macie
- B** Internet gateway
- C** AWS PrivateLink
- D** Amazon CloudFront

CORRECT ANSWER**Option C — AWS PrivateLink****EXPLANATION**

AWS PrivateLink enables private connectivity between a VPC and AWS services entirely within the AWS network — no internet gateway, no public IP addresses, and no exposure to internet traffic. This satisfies the regulatory requirement while still allowing the application to communicate with Amazon Bedrock. Amazon Macie is a data security service for S3, not a network connectivity solution. An internet gateway does the exact opposite — it enables internet access. Amazon CloudFront is a CDN that operates over the public internet.

Q2
4

AI/ML CONCEPTS

A law firm wants to build an AI application by using large language models (LLMs). The application will read legal documents and extract key points from the documents. Which solution meets these requirements?

- A Create a recommendation engine
- B Develop a summarization chatbot**
- C Develop a multi-language translation system
- D Build an automatic named entity recognition system

CORRECT ANSWER**Option B — Develop a summarization chatbot****EXPLANATION**

Summarization is exactly the task described — reading long legal documents and extracting key points is the definition of text summarization. LLMs excel at condensing lengthy contracts and case files into concise summaries of the most important information. A recommendation engine suggests items based on user preferences. Multi-language translation converts text between languages. Named entity recognition extracts specific entity types (names, dates) but cannot produce broad key points from a document.

Q2
5**RESPONSIBLE AI**

A medical company deployed a disease detection model on Amazon Bedrock. To comply with privacy policies, the company wants to prevent the model from including personal patient information in its responses and receive notification when policy violations occur. Which solution meets these requirements?

- A** Use Amazon Macie to scan the model's output for sensitive data and set up alerts for potential violations
- B** Configure AWS CloudTrail to monitor the model's responses and create alerts for any detected personal information
- C** Implement Amazon SageMaker Model Monitor to detect data drift and receive alerts when model quality degrades
- D** Use Guardrails for Amazon Bedrock to filter content. Set up Amazon CloudWatch alarms for notification of policy violations

CORRECT ANSWER

Option D — Use Guardrails for Amazon Bedrock to filter content. Set up Amazon CloudWatch alarms for notification of policy violations

EXPLANATION

Guardrails for Amazon Bedrock actively filters PII and sensitive patient data from model responses, preventing it from appearing in outputs. Amazon CloudWatch alarms provide real-time notifications when Guardrails detects and blocks a policy violation, satisfying both requirements. Amazon Macie scans data stored in S3 — it does not monitor real-time LLM responses. AWS CloudTrail logs API calls and audit events but cannot inspect or filter model response content. SageMaker Model Monitor detects data drift and quality degradation, not PII in Bedrock responses.

Q2
6**RESPONSIBLE AI**

A medical company is customizing a foundation model (FM) for diagnostic purposes. The company needs the model to be transparent and explainable to meet regulatory requirements. Which solution will meet these requirements?

- A Gather more data. Use Amazon Rekognition to add custom labels to the data
- B Generate simple metrics, reports, and examples by using Amazon SageMaker Clarify**
- C Configure the security and compliance by using Amazon Inspector
- D Encrypt and secure training data by using Amazon Macie

CORRECT ANSWER

Option B — Generate simple metrics, reports, and examples by using Amazon SageMaker Clarify

EXPLANATION

Amazon SageMaker Clarify is purpose-built for model transparency and explainability. It provides feature importance scores (showing which inputs drive predictions), bias detection reports, and SHAP-based explainability metrics. Amazon Rekognition with custom labels is a computer vision service — adding labels to data is a data preparation step, not an explainability solution. Amazon Inspector is a vulnerability assessment service for EC2 and containers, unrelated to model explainability. Amazon Macie discovers sensitive data in S3 — it addresses data privacy, not model explainability.

Q2
7

GENERATIVE AI

A research company implemented a chatbot using Amazon Bedrock. The chatbot searches for answers from a large database of research papers. After multiple prompt engineering attempts, the company notices that the FM performs poorly because of complex scientific terms. How can the company improve the performance?

- A Change the FM inference parameters
- B Clean the research paper data to remove complex scientific terms
- C Use domain adaptation fine-tuning to adapt the FM to complex scientific terms**
- D Use few-shot prompting to define how the FM can answer the questions

CORRECT ANSWER

Option C — Use domain adaptation fine-tuning to adapt the FM to complex scientific terms

EXPLANATION

Domain adaptation fine-tuning trains the foundation model on domain-specific data so the model learns the specialized vocabulary, terminology, and reasoning patterns of that field. Since prompt engineering has already been tried and failed due to the complexity of scientific terms, fine-tuning is the logical next step to deeply embed domain knowledge. Changing inference parameters like temperature controls output style, not the model's domain knowledge. Removing scientific terms would destroy the very information the chatbot needs to answer research questions. Few-shot prompting is still prompt engineering, which has already been exhausted.

Q2
8**AWS SECURITY**

A security company wants to ensure that only authorized users invoke Amazon Bedrock models. The company needs to identify any unauthorized access attempts to set appropriate IAM policies and roles. Which AWS service should the company use?

- A** AWS Audit Manager
- B** AWS CloudTrail
- C** Amazon Fraud Detector
- D** AWS Trusted Advisor

CORRECT ANSWER**Option B — AWS CloudTrail****EXPLANATION**

AWS CloudTrail records every API call made to AWS services including every attempt to invoke Amazon Bedrock models. It provides a complete audit log of who tried to access Bedrock, when, from where, and whether the attempt succeeded or was denied. Unauthorized access attempts appear as denied API calls. AWS Audit Manager assesses compliance against frameworks and is not designed for real-time unauthorized access detection. Amazon Fraud Detector detects fraudulent business transactions, not AWS infrastructure access attempts. AWS Trusted Advisor provides best-practice recommendations, not incident-level access monitoring.

Q2
9

RESPONSIBLE AI

A social media company wants to use a large language model (LLM) for content moderation. The company wants to evaluate the LLM outputs for bias and potential discrimination against specific groups. Which data source should the company use with the LEAST administrative effort?

A Benchmark datasets

B Content moderation guidelines

C User-generated content

D Moderation logs

CORRECT ANSWER

Option A — Benchmark datasets

EXPLANATION

Benchmark datasets are pre-built, standardized datasets specifically designed to evaluate model outputs for bias, fairness, and discrimination. They require no collection, cleaning, labeling, or curation — ready to use immediately. Content moderation guidelines are policy documents, not evaluation datasets — converting them requires substantial manual effort. User-generated content must be collected, filtered, labeled, and curated before use — high administrative overhead with privacy concerns. Moderation logs need to be extracted, cleaned, and anonymized, and also risk perpetuating existing human biases.

Q3
0

RESPONSIBLE AI

A student at a university is copying content from generative AI to write essays. Which challenge of responsible generative AI does this scenario represent?

- A Toxicity
- B Privacy
- C Hallucinations

D Plagiarism

CORRECT ANSWER

Option D — Plagiarism

EXPLANATION

This scenario directly represents plagiarism — the act of presenting AI-generated content as one's own original work without proper attribution. Toxicity refers to AI generating harmful or offensive content — the issue here is attribution, not the nature of the content. Privacy concerns AI exposing personal or sensitive data — no personal data is compromised here. Hallucinations refers to AI confidently generating factually incorrect information. While hallucinations could appear in the copied content, the challenge being illustrated is the act of misrepresenting authorship.

Q3
1**AWS SERVICES**

An AI company periodically evaluates its systems with the help of ISVs. The company needs to receive email notifications when an ISV's compliance reports become available. Which AWS service meets this requirement?

A AWS Audit Manager

B AWS Artifact

C AWS Data Exchange

D AWS Trusted Advisor

CORRECT ANSWER**Option B — AWS Artifact****EXPLANATION**

AWS Artifact is the AWS service for accessing compliance reports, certifications, and agreements including those from ISVs. It provides on-demand access to compliance documentation and allows companies to configure email notifications when new or updated compliance reports become available. AWS Audit Manager helps continuously audit AWS usage against frameworks — it generates internal audit evidence, not ISV compliance reports. AWS Data Exchange is a marketplace for subscribing to third-party data products, not compliance reports. AWS Trusted Advisor provides best-practice recommendations for your own AWS environment.

Q3
2

AI/ML CONCEPTS

An AI practitioner has a database of animal photos. The AI practitioner wants to automatically identify and categorize the animals in the photos without manual human effort. Which strategy meets these requirements?

- A Anomaly detection
- B Inpainting
- C Named entity recognition

D Object detection

CORRECT ANSWER

Option D — Object detection

EXPLANATION

Object detection is specifically designed to automatically identify and locate objects (in this case, animals) within images and classify them into categories, all without manual human effort. It does both jobs: finding where the animal is and identifying what animal it is. Anomaly detection identifies unusual patterns, not for categorizing objects in images. Inpainting fills in missing parts of an image and is used for restoration, not classification. Named entity recognition is an NLP technique for extracting entities from text, not images.

Q3
3

ML CONCEPTS

An AI practitioner has built a deep learning model to classify types of materials in images and wants to measure model performance. Which metric will help evaluate the performance of the model?

- A Correlation matrix
- B R2 score
- C Mean squared error (MSE)

D Confusion matrix

CORRECT ANSWER

Option D — Confusion matrix

EXPLANATION

A confusion matrix is the ideal performance evaluation tool for a classification model. It shows a breakdown of correct and incorrect predictions across every class, making it easy to see overall accuracy and where the model is making mistakes. A correlation matrix shows statistical relationships between variables in a dataset — a data exploration tool, not a model performance metric. R-squared measures how well a regression model explains variance in continuous numerical output — meaningless for classification. MSE measures the average squared difference between predicted and actual numerical values — a regression metric, not suitable for classification.

Q3
4**RESPONSIBLE AI**

An AI practitioner is building a model to generate images of humans in various professions. The AI practitioner discovered that the input data is biased and specific attributes affect the image generation and create bias in the model. Which technique will solve the problem?

A Retrieval Augmented Generation (RAG)

B Data augmentation for imbalanced classes

C Model monitoring for class distribution

D Watermark detection for images

CORRECT ANSWER

Option B — Data augmentation for imbalanced classes

EXPLANATION

Data augmentation for imbalanced classes directly addresses the root cause — biased input data. If the training data over-represents certain demographics in specific professions, data augmentation can synthetically generate or oversample underrepresented groups to create a more balanced dataset, reducing bias in generated images. RAG is designed for text generation accuracy, not for fixing demographic bias in image generation training data. Model monitoring tracks bias after deployment but does not solve the problem. Watermark detection identifies AI-generated content for copyright purposes — completely unrelated to training data bias.

Q3
5

AWS SERVICES

An AI practitioner is using an Amazon Bedrock base model to summarize session chats from the customer service department. The AI practitioner wants to store invocation logs to monitor model input and output data. Which strategy should the AI practitioner use?

- A Configure AWS CloudTrail as the logs destination for the model
- B Configure model invocation logging in Amazon EventBridge
- C Configure AWS Audit Manager as the logs destination for the model
- D Enable invocation logging in Amazon Bedrock**

CORRECT ANSWER

Option D — Enable invocation logging in Amazon Bedrock

EXPLANATION

Amazon Bedrock has a built-in invocation logging feature that captures the full input prompts and output responses for every model invocation. Logs can be sent to Amazon S3 or Amazon CloudWatch Logs. AWS CloudTrail logs API-level calls (metadata like who called the API, when, and with what parameters) but does NOT capture the actual content of the model's input prompts or output responses. Amazon EventBridge is an event bus for routing events — not a logging destination for model invocation content. AWS Audit Manager collects evidence for compliance assessments and is not designed for real-time model I/O data.

Q3
6

GENERATIVE AI

An AI practitioner wants to use a foundation model (FM) to design a search application. The search application must handle queries that have text and images. Which type of FM should the AI practitioner use?

A Multi-modal embedding model

B Image generation model

C Multi-modal generation model

D Text embedding model

CORRECT ANSWER

Option A — Multi-modal embedding model

EXPLANATION

A multi-modal embedding model converts both text and images into a shared vector embedding space, enabling the search application to compare and retrieve results across different modalities. When a user submits a query with text and images, the model embeds both into vectors for similarity search. An image generation model generates new images — it creates content, not powers search. A multi-modal generation model generates text or image outputs from multi-modal inputs — generation is about creating content, not indexing and retrieving it. A text embedding model cannot process image inputs.

Q3
7**AWS SECURITY**

How can companies use large language models (LLMs) securely on Amazon Bedrock?

- A** Enable Amazon Bedrock automatic model evaluation jobs
- B** Use Amazon CloudWatch Logs to make models explainable and to monitor for bias
- C** Enable AWS Audit Manager for automatic model evaluation jobs
- D** Design clear and specific prompts. Configure AWS Identity and Access Management (IAM) roles and policies by using least privilege access

CORRECT ANSWER

Option D — Design clear and specific prompts. Configure AWS Identity and Access Management (IAM) roles and policies by using least privilege access

EXPLANATION

This answer addresses both dimensions of secure LLM use: clear and specific prompts reduce the risk of prompt injection attacks and unintended model behavior, while IAM least privilege ensures only authorized users can invoke Bedrock models. Automatic model evaluation jobs assess model quality and performance, not security. CloudWatch Logs is a monitoring service — it does not provide model explainability or bias detection (SageMaker Clarify does those). AWS Audit Manager collects compliance evidence but does not run model evaluation jobs or enforce secure usage patterns.

Q3
8

AWS SERVICES

Which AWS service or feature can help an AI development team quickly deploy and consume a foundation model (FM) within the team's VPC?

- A Amazon Personalize
- B Amazon SageMaker JumpStart**
- C PartyRock, an Amazon Bedrock Playground
- D Amazon SageMaker endpoints

CORRECT ANSWER**Option B — Amazon SageMaker JumpStart****EXPLANATION**

Amazon SageMaker JumpStart provides a model hub with hundreds of pre-trained foundation models deployable with one click directly into a SageMaker endpoint within the team's VPC. It is specifically designed for rapid deployment without building infrastructure from scratch. Amazon Personalize is a recommendation system service, not a platform for deploying foundation models. PartyRock is a public web-based playground for prototyping — not a VPC-deployable service for production use. SageMaker endpoints host deployed models but require manual setup. JumpStart uses endpoints under the hood but adds the model hub and one-click deployment experience.

Q3
9

AWS SERVICES

Which feature of Amazon OpenSearch Service gives companies the ability to build vector database applications?

A Ability to perform real-time analysis on streaming data

B Integration with Amazon S3 for object storage

C Scalable index management and nearest neighbor search capability

D Support for geospatial indexing and queries

CORRECT ANSWER

Option C — Scalable index management and nearest neighbor search capability

EXPLANATION

Amazon OpenSearch Service supports k-Nearest Neighbor (k-NN) search, which is the core capability that makes it function as a vector database. It can store high-dimensional vector embeddings and efficiently retrieve the most similar vectors to a query vector — the fundamental operation powering semantic search, recommendation engines, and RAG applications. Real-time analysis on streaming data is a general analytics feature unrelated to vector database functionality. S3 integration enables querying data from S3 but has nothing to do with vector similarity search. Geospatial indexing supports location-based searches — a completely separate capability from vector embedding search.

Q4
0

AWS SERVICES

Which functionality does Amazon SageMaker Clarify provide?

- A Documents critical details about ML models
- B Identifies potential bias during data preparation**
- C Integrates a Retrieval Augmented Generation (RAG) workflow
- D Monitors the quality of ML models in production

CORRECT ANSWER

Option B — Identifies potential bias during data preparation

EXPLANATION

Amazon SageMaker Clarify is specifically designed to detect bias in datasets and model predictions, and provide explainability for model outputs. During data preparation, Clarify can analyze datasets to identify imbalances and biases across sensitive attributes before a model is trained. Documenting critical details about ML models is the function of SageMaker Model Cards. Integrating a RAG workflow is the function of Amazon Bedrock Knowledge Bases. Monitoring the quality of ML models in production is the function of SageMaker Model Monitor.

Q4
1

ML CONCEPTS

Which metric measures the runtime efficiency of operating AI models?

A Customer satisfaction score (CSAT)

B Average response time

C Number of training instances

D Training time for each epoch

CORRECT ANSWER

Option B — Average response time

EXPLANATION

Average response time measures how long an AI model takes to process a request and return a result during runtime operation (inference). It directly reflects the efficiency of the deployed model in production. CSAT measures how satisfied users are with a product — a business metric, not a technical runtime efficiency measure. Number of training instances refers to the size of the training dataset, not a runtime efficiency metric. Training time per epoch measures efficiency during model training, not during runtime inference.

Q4
2

ML CONCEPTS

A company has built an image classification model to predict plant diseases from photos of plant leaves. The company wants to evaluate how many images the model classified correctly. Which evaluation metric should the company use?

- A Learning rate
- B Root mean squared error (RMSE)
- C Accuracy**
- D R-squared score

CORRECT ANSWER

Option C — Accuracy

EXPLANATION

Accuracy is the most direct metric for measuring how many predictions a classification model got right: $\text{Accuracy} = \text{Correct Predictions} / \text{Total Predictions} \times 100\%$. Since the company explicitly wants to know how many images were classified correctly, accuracy answers that directly. Learning rate is a training hyperparameter that controls weight updates — not an evaluation metric. RMSE measures average magnitude of errors between predicted and actual continuous values — a regression metric, not classification. R-squared measures variance explained by a regression model — also inappropriate for image classification.

Q4
3

AWS SERVICES

A company has developed an ML model for image classification and wants to deploy it to production for a web application. The company needs to host the model and serve predictions without managing any underlying infrastructure. Which solution will meet these requirements?

A Use Amazon API Gateway to host the model and serve predictions

B Use Amazon CloudFront to deploy the model

C Use AWS Batch to host the model and serve predictions

D Use Amazon SageMaker Serverless Inference to deploy the model

CORRECT ANSWER

Option D — Use Amazon SageMaker Serverless Inference to deploy the model

EXPLANATION

Amazon SageMaker Serverless Inference is specifically designed to host ML models and serve predictions without any infrastructure management. AWS automatically provisions, scales, and manages all underlying compute. Amazon API Gateway creates and manages REST/HTTP APIs but cannot host or serve an ML model on its own. Amazon CloudFront is a CDN for caching and distributing static content — no ability to run ML model inference. AWS Batch runs large-scale batch computing jobs and requires infrastructure configuration — designed for offline batch processing, not real-time web predictions.

Q4
4

RESPONSIBLE AI

A company has installed a security camera. The company uses an ML model to evaluate footage for potential thefts. The company has discovered that the model disproportionately flags people who are members of a specific ethnic group. Which type of bias is affecting the model output?

A Measurement bias

B Confirmation bias

C Observer bias

D Sampling bias

CORRECT ANSWER

Option D — Sampling bias

EXPLANATION

Sampling bias occurs when the training data is not representative of the real-world population the model will encounter. The security camera model was likely trained on a dataset that over-represented a specific ethnic group in theft-related incidents, causing the model to learn a skewed association. Measurement bias occurs when the data collection method systematically produces inaccurate values — the issue here is who is in the data, not how it was measured. Confirmation bias is a human cognitive bias where people favor confirming information — a psychology concept, not an ML data composition issue. Observer bias occurs when the person collecting or labeling data unconsciously influences results.

Q4
5

ML CONCEPTS

A company has petabytes of unlabeled customer data to use for an advertisement campaign. The company wants to classify its customers into tiers to advertise and promote the company's products. Which methodology should the company use?

- A Reinforcement learning
- B Supervised learning
- C Reinforcement learning from human feedback (RLHF)

D Unsupervised learning

CORRECT ANSWER

Option D — Unsupervised learning

EXPLANATION

Unsupervised learning is correct because the data is unlabeled and the goal is to discover natural groupings (customer tiers) within the data without predefined categories. Clustering algorithms will automatically group customers based on shared characteristics. Reinforcement learning trains an agent to make sequential decisions — it does not apply to customer segmentation from static data. Supervised learning requires labeled training data, which the company explicitly does not have. RLHF is used to align LLMs with human preferences using human ratings — entirely irrelevant to customer segmentation.

Q4
6

GENERATIVE AI

A company has terabytes of data in a database. The company wants to build an AI-based application that can build a SQL query from input text that employees provide. Employees have minimal technology experience. Which solution meets these requirements?

A Support vector machine

B WaveNet

C Generative pre-trained transformers (GPT)

D Residual neural network

CORRECT ANSWER

Option C — Generative pre-trained transformers (GPT)

EXPLANATION

GPT models are large language models trained on vast amounts of text including code and SQL, making them highly capable of understanding natural language input and generating accurate SQL queries. This is a classic text-to-SQL task that LLMs excel at. Support Vector Machine is a classification/regression algorithm with no capability to understand natural language or generate SQL queries. WaveNet is a deep learning model for audio and speech synthesis — not text-to-SQL or NLP. Residual Neural Network (ResNet) is a deep convolutional architecture for image recognition — it cannot process natural language or generate SQL.

Q4
7

AWS SERVICES

A company is building a contact center application and wants to analyze and extract key information from the audio of customer calls. Which solution meets these requirements?

- A Build a conversational chatbot by using Amazon Lex
- B Create classification labels by using Amazon Comprehend
- C Extract information from call recordings by using Amazon SageMaker Model Monitor

D Transcribe call recordings by using Amazon Transcribe

CORRECT ANSWER

Option D — Transcribe call recordings by using Amazon Transcribe

EXPLANATION

Amazon Transcribe converts audio recordings of customer calls into text, which is the essential first step for analyzing and extracting key information from spoken conversations. Amazon Lex builds conversational chatbots for new interactions — it does not analyze existing recorded audio files. Amazon Comprehend analyzes text for sentiment and entities — while useful after transcription, it cannot process raw audio directly. SageMaker Model Monitor monitors deployed ML models for data drift — no capability to process audio or extract information from call recordings.

Q4
8

ML CONCEPTS

A company is building a customer service chatbot. The company wants the chatbot to improve its responses by learning from past interactions and online resources. Which AI learning strategy provides this self-improvement capability?

- A Supervised learning with a manually curated dataset of good and bad responses
- B Reinforcement learning with rewards for positive customer feedback**
- C Supervised learning with a continuously updated FAQ database
- D Unsupervised learning to find clusters of similar customer inquiries

CORRECT ANSWER

Option B — Reinforcement learning with rewards for positive customer feedback

EXPLANATION

Reinforcement learning enables the chatbot to continuously self-improve by learning from the outcomes of its own interactions. When customers respond positively, that response receives a reward signal, reinforcing the behavior. This creates a continuous improvement loop driven by real interaction data. Supervised learning with manually curated data requires humans to label examples every time — not self-improvement. Supervised learning with a FAQ database relies on humans maintaining the database. Unsupervised learning identifies patterns in questions but does not optimize response quality over time.

Q4
9

AWS SERVICES

A company is building a solution to generate images for protective eyewear. The solution must have high accuracy and must minimize the risk of incorrect annotations. Which solution will meet these requirements?

- A Data augmentation by using an Amazon Bedrock knowledge base
- B Human-in-the-loop validation by using Amazon SageMaker Ground Truth Plus**
- C Image recognition by using Amazon Rekognition
- D Data summarization by using Amazon QuickSight Q

CORRECT ANSWER

Option B — Human-in-the-loop validation by using Amazon SageMaker Ground Truth Plus

EXPLANATION

Amazon SageMaker Ground Truth Plus provides a human-in-the-loop labeling and annotation workflow where trained human reviewers validate and correct annotations, directly minimizing the risk of incorrect annotations. For a specialized domain like protective eyewear, human expert review ensures high accuracy. Data augmentation using Bedrock knowledge base generates variations of data but does not validate annotations. Amazon Rekognition can analyze images but cannot provide specialized annotation validation for domain-specific eyewear imagery with minimal error risk. Amazon QuickSight Q is a business intelligence tool for data visualization — not image annotation.

Q5
0

ML CONCEPTS

A company is building an application that needs to generate synthetic data based on existing data. Which type of model can the company use to meet this requirement?

- A XGBoost
- B Residual neural network
- C Generative adversarial network (GAN)**
- D WaveNet

CORRECT ANSWER**Option C — Generative adversarial network (GAN)****EXPLANATION**

A GAN is specifically designed to generate realistic synthetic data that resembles existing training data. The Generator creates synthetic samples while the Discriminator evaluates whether samples are real or generated. Through adversarial training, the generator continuously improves until it produces synthetic data indistinguishable from real data. XGBoost is a gradient boosting algorithm for classification and regression — it makes predictions, not generate new data. ResNet is a deep convolutional architecture primarily for image classification. WaveNet is a generative model specifically for audio and speech synthesis, not general synthetic data generation.

Q5
1**AWS SERVICES**

A company is building an ML model to analyze archived data. The company must perform inference on large datasets that are multiple GBs in size. The company does not need to access the model predictions immediately. Which Amazon SageMaker inference option will meet these requirements?

- A** Serverless inference
- B** Asynchronous inference
- C** Real-time inference

D Batch transform

CORRECT ANSWER

Option D — Batch transform

EXPLANATION

Amazon SageMaker Batch Transform is purpose-built for running inference on large datasets (multiple GBs) where results are not needed immediately. It processes entire datasets as an offline job, writes predictions to Amazon S3, and shuts down compute when complete. Serverless inference is designed for small, intermittent requests with strict payload size limits — not for large offline datasets. Asynchronous inference handles large individual payloads (up to 1 GB) with near real-time results — designed for single large requests, not entire multi-GB archived datasets. Real-time inference is designed for low-latency, immediate predictions on small payloads (up to 6 MB).

Q5
2

ML CONCEPTS

A company is developing a new model to predict the prices of specific items. The model performed well on the training dataset but when deployed to production, the model's performance decreased significantly. What should the company do to mitigate this problem?

- A Reduce the volume of data that is used in training
- B Increase the volume of data that is used in training**
- C Add hyperparameters to the model
- D Increase the model training time

CORRECT ANSWER

Option B — Increase the volume of data that is used in training

EXPLANATION

The scenario describes overfitting — the model memorized the training data and fails to generalize to new real-world data. The most effective remedy is to increase the volume and diversity of training data, exposing the model to more varied examples so it learns general patterns rather than memorizing specific ones. Reducing training data would make overfitting worse by giving the model even fewer patterns to generalize from. Hyperparameters are configuration settings — adding them does not solve overfitting. Training longer on the same dataset deepens memorization, making overfitting more severe.

Q5
3

GENERATIVE AI

A company is implementing the Amazon Titan foundation model using Amazon Bedrock. The company needs to supplement the model by using relevant data from the company's private data sources. Which solution will meet this requirement?

- A Choose a lower temperature value
- B Enable model invocation logging
- C Create an Amazon Bedrock knowledge base**
- D Use a different FM

CORRECT ANSWER

Option C — Create an Amazon Bedrock knowledge base

EXPLANATION

An Amazon Bedrock knowledge base implements RAG, connecting the foundation model to the company's private data sources. When a user submits a query, the knowledge base retrieves the most relevant chunks from the private data and passes them as context to the model. Choosing a lower temperature adjusts output randomness — it does not add new knowledge or connect the model to private data. Enabling invocation logging records inputs and outputs for monitoring — not a data supplementation mechanism. Using a different FM doesn't solve the problem — no base FM has access to the company's private data regardless of which model is chosen.

Q5
4

ML CONCEPTS

A company is training a foundation model (FM). The company wants to increase the accuracy of the model up to a specific acceptance level. Which solution will meet these requirements?

- A** Increase the epochs
- B Increase the temperature parameter
- C Decrease the batch size
- D Decrease the epochs

CORRECT ANSWER**Option A — Increase the epochs****EXPLANATION**

An epoch is one complete pass through the entire training dataset. Increasing the number of epochs gives the model more opportunities to learn from training data, progressively refining its weights and improving accuracy until the desired level is reached. Temperature is an inference parameter that controls output randomness — not a training parameter with no effect on accuracy during training. Simply decreasing batch size does not reliably increase accuracy to a target level. Decreasing epochs means fewer training passes, which would reduce accuracy, not increase it. Note: increasing epochs beyond the optimal point can cause overfitting, so early stopping should be used.

Q5
5

GENERATIVE AI

A company is using a pre-trained large language model (LLM) to build a chatbot for product recommendations. The company needs the LLM outputs to be short and written in a specific language. Which solution will align the LLM response quality with the company's expectations?

- A Increase the Top K value
- B Adjust the prompt**
- C Increase the temperature
- D Choose an LLM of a different size

CORRECT ANSWER

Option B — Adjust the prompt

EXPLANATION

Prompt engineering is the most direct and effective solution for controlling both the length and language of LLM outputs. By explicitly instructing the model in the prompt (e.g., 'Respond only in French. Keep your response under 3 sentences'), the model follows these instructions. Increasing Top K adds variety and randomness to outputs but does nothing to enforce response length or language. Increasing temperature raises output randomness, working against the goal of consistent short responses. Model size affects general capability, not the language or length of output.

Q5
6

GENERATIVE AI

A company is using an Amazon Bedrock base model to summarize documents for an internal use case. The company trained a custom model to improve summarization quality. Which action must the company take to use the custom model through Amazon Bedrock?

- A Purchase Provisioned Throughput for the custom model**
- B Register the model with the Amazon SageMaker Model Registry
- C Grant access to the custom model in Amazon Bedrock
- D Deploy the custom model in an Amazon SageMaker endpoint for real-time inference

CORRECT ANSWER

Option A — Purchase Provisioned Throughput for the custom model

EXPLANATION

When a company fine-tunes or customizes a foundation model in Amazon Bedrock, the resulting custom model cannot be used with On-Demand pricing. To invoke a custom model through Amazon Bedrock, the company must purchase Provisioned Throughput — which reserves dedicated capacity and makes it available for inference. SageMaker Model Registry catalogs models within SageMaker pipelines and has no role in making a Bedrock custom model available. Simply granting IAM access alone is not sufficient without Provisioned Throughput. Deploying to a SageMaker endpoint is for SageMaker-hosted models — Bedrock custom models are invoked through the Bedrock API.

Q5
7

ML CONCEPTS

A company is using domain-specific models. The company wants to avoid creating new models from scratch. The company instead wants to adapt pre-trained models to create models for new, related tasks. Which ML strategy meets these requirements?

- A Use unsupervised learning
- B Decrease the number of epochs
- C Use transfer learning**
- D Increase the number of epochs

CORRECT ANSWER

Option C — Use transfer learning

EXPLANATION

Transfer learning is exactly the strategy described — taking a pre-trained model that has already learned knowledge from a large dataset and adapting it to a new, related task without training from scratch. The pre-trained model's learned weights serve as a starting point, dramatically reducing data, compute, and time required. Unsupervised learning is a methodology for finding patterns in unlabeled data, not a strategy for adapting existing pre-trained models. Decreasing epochs results in less learning and lower model performance — unrelated to reusing pre-trained models. Increasing epochs trains longer from the current starting point but does not leverage pre-existing model knowledge.

Q5
8

RESPONSIBLE AI

A company is using the Generative AI Security Scoping Matrix to assess security responsibilities. The company has identified four different solution scopes. Which solution scope gives the company the MOST ownership of security responsibilities?

- A Using a third-party enterprise application that has embedded generative AI features
- B Building and training a generative AI model from scratch using specific data that a customer owns**
- C Refining an existing third-party generative AI FM by fine-tuning the model using business-specific data
- D Building an application by using an existing third-party generative AI foundation model (FM)

CORRECT ANSWER

Option B — Building and training a generative AI model from scratch using specific data that a customer owns

EXPLANATION

Building a generative AI model completely from scratch places maximum security responsibility on the company. Every layer — data collection, data security, model architecture, training infrastructure, model storage, deployment, access control, and monitoring — is entirely the company's responsibility. Using a third-party enterprise application with embedded AI gives the company the least security ownership (vendor manages most controls). Building an app on a third-party FM means the company owns the application layer but relies on the FM provider for model security. Fine-tuning an existing FM places medium ownership — the company owns the fine-tuning data and adapted model but the base model infrastructure remains the vendor's responsibility.

Q5
9

AWS SERVICES

A company wants to monitor its Amazon Bedrock application for unusual patterns and detect potential security threats in real time. The company wants to receive automated alerts when suspicious behavior is detected. Which combination of AWS services meets these requirements?

- A AWS Artifact and Amazon SNS
- B Amazon CloudWatch and AWS CloudTrail**
- C Amazon Macie and AWS Audit Manager
- D AWS Trusted Advisor and Amazon Inspector

CORRECT ANSWER

Option B — Amazon CloudWatch and AWS CloudTrail

EXPLANATION

CloudTrail logs all API calls (who invoked Bedrock, when, from where, success/failure) and CloudWatch analyzes those logs, detects anomalous patterns, and triggers automated alerts via CloudWatch Alarms and SNS. Together they form the standard AWS security monitoring pattern. Macie only scans S3 data. Artifact is for compliance reports. Trusted Advisor provides recommendations, not real-time alerting.

Q6
0

RESPONSIBLE AI

A company has a generative AI chatbot that sometimes produces factually incorrect information confidently. Which challenge of responsible AI does this represent, and what is the most appropriate mitigation?

- A Toxicity — implement Guardrails for Amazon Bedrock to filter harmful content
- B Hallucinations — implement RAG to ground responses in verified, retrieved facts**
- C Plagiarism — add watermark detection to identify AI-generated content
- D Bias — use Amazon SageMaker Clarify to detect bias in model outputs

CORRECT ANSWER

Option B — Hallucinations — implement RAG to ground responses in verified, retrieved facts

EXPLANATION

Confidently generating false information is the definition of hallucinations. RAG mitigates this by grounding the model's responses in retrieved, verified facts from a knowledge base — the model can only answer based on what it retrieves. Guardrails filters content but doesn't fix factual accuracy. Clarify addresses bias, not factual errors. Watermark detection identifies AI-generated content but doesn't fix the underlying hallucination problem.

Q6
1

ML CONCEPTS

A hospital wants to predict whether a patient will be readmitted within 30 days of discharge. The hospital has 5 years of labeled patient records with readmission outcomes. Which ML approach is MOST appropriate?

- A Unsupervised learning using clustering to group similar patients
- B Reinforcement learning to optimize hospital discharge decisions
- C Supervised learning using binary classification with labeled historical outcomes**
- D Generative AI to create synthetic patient records for training

CORRECT ANSWER

Option C — Supervised learning using binary classification with labeled historical outcomes

EXPLANATION

This is a classic supervised learning binary classification problem — labeled data (readmitted: yes/no) exists, and the goal is to predict a known categorical outcome. Unsupervised learning is for unlabeled data — this hospital has labeled data. Reinforcement learning optimizes sequential decisions through rewards, not static prediction from historical records. Synthetic data generation doesn't solve the prediction problem — the hospital already has real labeled data.

Q6
2

ML CONCEPTS

A machine learning model for fraud detection achieves 99% accuracy on a dataset where 99% of transactions are legitimate. What is the MOST significant concern with this result?

- A The model is overfitting to the training data
- B The accuracy metric is misleading due to severe class imbalance**
- C The model needs more training epochs to improve
- D The temperature parameter needs to be decreased

CORRECT ANSWER

Option B — The accuracy metric is misleading due to severe class imbalance

EXPLANATION

This is the classic class imbalance problem. A model that predicts 'legitimate' for every single transaction would achieve 99% accuracy but catch zero fraud — making it completely useless. When classes are severely imbalanced, accuracy is a misleading metric. Better metrics are Precision, Recall, F1 score, and AUC-ROC, which evaluate performance on the minority (fraud) class specifically. There is no indication of overfitting — the concern is the metric choice. More epochs won't fix metric selection. Temperature is an inference parameter, not relevant here.

Q6
3

AWS SERVICES

A data science team wants to make their ML features available to multiple teams across the organization, ensuring everyone uses the same consistent feature definitions for both training and real-time inference. Which AWS service should they use?

- A Amazon SageMaker Data Wrangler
- B Amazon SageMaker Feature Store**
- C Amazon SageMaker Model Registry
- D Amazon SageMaker Pipelines

CORRECT ANSWER

Option B — Amazon SageMaker Feature Store

EXPLANATION

SageMaker Feature Store is the centralized repository for storing, versioning, and sharing ML features across teams. It has an online store (low-latency for real-time inference) and offline store (for batch training), ensuring the same feature definitions are used consistently in both contexts. Data Wrangler is for data preparation and transformation. Model Registry catalogs and versions trained ML models. Pipelines automates ML workflows — none of these provide cross-team feature sharing.

Q6
4

GENERATIVE AI

A company needs to process 10,000 product images to classify them into categories for an e-commerce catalog update. The results are needed within a few hours but not immediately. Which SageMaker inference option is MOST appropriate?

- A Real-time inference with an auto-scaling endpoint
- B Serverless inference for on-demand processing
- C Batch Transform to process all images as an offline job**
- D Asynchronous inference for near real-time results

CORRECT ANSWER

Option C — Batch Transform to process all images as an offline job

EXPLANATION

Batch Transform is designed for processing large datasets offline where results are not needed immediately. Processing 10,000 images as a bulk offline job and writing results to S3 is the most cost-effective approach. Real-time inference keeps an expensive persistent endpoint running unnecessarily. Serverless has payload size limits and is designed for individual on-demand requests. Async is designed for single large requests, not bulk dataset processing across thousands of images.

Q6
5

AWS SERVICES

A company's ML model deployed on SageMaker was performing well initially but its prediction accuracy has been declining over the past 3 months as market conditions have changed. Which AWS service should the company use to detect and alert on this issue automatically?

- A Amazon SageMaker Clarify to detect bias in model predictions
- B Amazon SageMaker Model Monitor to detect data drift and model quality degradation**
- C AWS CloudTrail to monitor API calls to the SageMaker endpoint
- D Amazon Bedrock Guardrails to filter low-quality model outputs

CORRECT ANSWER

Option B — Amazon SageMaker Model Monitor to detect data drift and model quality degradation

EXPLANATION

Declining accuracy due to changing real-world conditions is called model drift or data drift — the statistical properties of the input data have changed from what the model was trained on. SageMaker Model Monitor continuously monitors production endpoints for data drift, model quality degradation, and bias drift, sending CloudWatch alerts when thresholds are breached. Clarify is for pre-deployment bias and explainability analysis, not ongoing production monitoring. CloudTrail logs API calls (metadata), not model prediction quality. Guardrails filters LLM content in Bedrock — not applicable to SageMaker model quality monitoring.

Q6
6

GENERATIVE AI

A startup is building an MVP generative AI application and needs to choose a foundation model on Amazon Bedrock. They have variable and unpredictable traffic, a limited budget, and do not want any long-term commitments. Which combination of choices is **MOST** appropriate?

- A Choose the largest available model with Provisioned Throughput for performance guarantees
- B Choose an appropriately sized model with On-Demand pricing for flexibility**
- C Fine-tune a model first, then deploy with Provisioned Throughput
- D Use SageMaker JumpStart with a reserved instance for cost savings

CORRECT ANSWER

Option B — Choose an appropriately sized model with On-Demand pricing for flexibility

EXPLANATION

For an MVP with variable traffic and budget constraints, On-Demand pricing means no commitment and payment only for what is used — perfect for unpredictable traffic. Choosing an appropriately sized (not necessarily the largest) model balances capability and cost. Provisioned Throughput requires upfront commitment — wrong for MVPs with unpredictable traffic. Fine-tuning plus Provisioned Throughput is expensive, technically complex, and premature at MVP stage. SageMaker JumpStart with reserved instances also involves commitment and is more appropriate for production workloads.

Q6
7

GENERATIVE AI

A company wants to use Amazon Bedrock to build an AI assistant that can autonomously search the web, query an internal database, and send email notifications — completing a multi-step workflow without human intervention for each step. Which Amazon Bedrock feature should the company use?

- A Amazon Bedrock Knowledge Bases for retrieving information
- B Amazon Bedrock Guardrails for controlling model behavior
- C Amazon Bedrock Agents for orchestrating multi-step autonomous workflows**
- D Amazon Bedrock Model Evaluation for testing the assistant's capabilities

CORRECT ANSWER

Option C — Amazon Bedrock Agents for orchestrating multi-step autonomous workflows

EXPLANATION

Amazon Bedrock Agents are designed specifically for multi-step autonomous task execution. Agents can call external APIs, query databases, search the web, trigger Lambda functions, and chain multiple actions together to complete complex workflows without requiring human approval for each step. Knowledge Bases retrieves information for RAG but doesn't orchestrate multi-step actions. Guardrails controls content filtering, not workflow execution. Model Evaluation assesses model quality — not for building autonomous workflows.

Q6
8

RESPONSIBLE AI

An AI model used for hiring decisions shows that it recommends significantly fewer candidates from one geographic region compared to others with similar qualifications. Which type of bias is MOST likely causing this issue?

- A Confirmation bias from hiring managers reviewing the outputs
- B Measurement bias from inaccurate qualification assessment tools
- C Historical bias embedded in the training data from past hiring decisions**
- D Observer bias from the team that built the model

CORRECT ANSWER

Option C — Historical bias embedded in the training data from past hiring decisions

EXPLANATION

When an AI hiring model reflects geographic or demographic disparities, the most likely cause is historical bias embedded in the training data. The model learned from past hiring decisions that were themselves biased. If training data shows fewer hires from a region historically, the model perpetuates that pattern. This is why diverse, representative training data and bias audits using SageMaker Clarify are essential for hiring AI systems. Confirmation bias is a human cognitive issue. Measurement bias involves inaccurate measurement tools. Observer bias involves unconscious influence during data collection.

Q6
9

AWS SERVICES

A company has sensitive ML training data stored in Amazon S3 encrypted with AWS KMS customer managed keys (CMKs). The company needs to train a SageMaker model using this data while maintaining encryption. What must the SageMaker execution role have?

- A S3 public access permissions to read the training data
- B Amazon Macie permissions to scan the data before training
- C KMS permissions to use the CMK for decryption, in addition to S3 read permissions**
- D AWS Audit Manager permissions to log the training job

CORRECT ANSWER

Option C — KMS permissions to use the CMK for decryption, in addition to S3 read permissions

EXPLANATION

When training data is encrypted with a KMS Customer Managed Key (CMK), the SageMaker execution role needs two permissions: (1) `s3:GetObject` permission to access the S3 objects, AND (2) `kms:Decrypt` permissions on the specific CMK. Without both, the training job will fail with an access denied error. Public access is a serious security violation. Macie scans S3 for sensitive data but has no role in granting training access. Audit Manager collects compliance evidence and is unrelated to training job execution permissions.

Q7
0

GENERATIVE AI

A company is implementing a multi-turn customer service chatbot using Amazon Bedrock. After several conversation turns, the chatbot starts to forget earlier context from the same conversation. What is the MOST likely cause?

- A The temperature parameter is set too high
- B The chatbot has exceeded the model's context window limit**
- C The model requires fine-tuning on customer service data
- D Guardrails are filtering earlier conversation turns

CORRECT ANSWER

Option B — The chatbot has exceeded the model's context window limit

EXPLANATION

LLMs have no persistent memory between API calls. The entire conversation history must be sent with each request. If the conversation exceeds the model's context window (token limit), earlier turns get truncated and forgotten. The fix is to use a model with a larger context window, implement a summarization strategy for older turns, or use conversation memory management. High temperature affects output randomness, not memory. Fine-tuning addresses domain knowledge, not conversation memory. Guardrails filter harmful content, not conversational context.

Q7
1**AWS SERVICES**

A company wants to build a model that can identify whether a product image shows a defect **AND** describe the defect in natural language. Which type of model architecture meets these requirements?

- A** A text embedding model for semantic search
- B** A multi-modal generation model that processes images and generates text
- C** A convolutional neural network (CNN) for image classification
- D** A recurrent neural network (RNN) for sequence processing

CORRECT ANSWER

Option B — A multi-modal generation model that processes images and generates text

EXPLANATION

This requires a model that can both understand visual input (image with defect) **AND** generate natural language output (description of the defect). Multi-modal generation models take images as input and generate text descriptions. A text embedding model converts text to vectors for search — it cannot process images or generate descriptions. A standard CNN classifies images but cannot generate descriptive natural language text. RNNs process sequential data like time series or text — they cannot process images as primary input.

Q7
2

ML CONCEPTS

A company wants to use large language models (LLMs) to generate marketing content. After testing, they find the outputs are too similar and repetitive across different campaign ideas. Which adjustment would **MOST** effectively increase the variety and creativity of outputs?

- A Decrease the context window size
- B Switch to a smaller language model
- C Increase the temperature parameter**
- D Enable Guardrails for Amazon Bedrock

CORRECT ANSWER**Option C — Increase the temperature parameter****EXPLANATION**

Increasing the temperature parameter makes the model's output more random and creative by sampling from a broader distribution of token probabilities. A higher temperature means the model is less likely to always pick the most probable next token, resulting in more varied, diverse, and creative outputs — perfect for generating different marketing campaign ideas. Decreasing context window size reduces how much information the model considers, not output creativity. A smaller model generally has less capability, not more creativity. Guardrails filters harmful or off-topic content — it has no effect on output diversity or creativity.

Q7
3

AWS SERVICES

A company wants to automatically detect when the inputs to their deployed SageMaker model begin to differ significantly from the data the model was trained on. Which SageMaker feature addresses this specific need?

- A SageMaker Clarify — to detect pre-training bias
- B SageMaker Data Wrangler — to re-process incoming data
- C SageMaker Model Monitor — Data Quality Monitor to detect input data drift**
- D SageMaker Ground Truth — to re-label incoming production data

CORRECT ANSWER

Option C — SageMaker Model Monitor — Data Quality Monitor to detect input data drift

EXPLANATION

SageMaker Model Monitor's Data Quality Monitor continuously compares the statistical properties of production input data against a baseline collected from training data. When the distribution of incoming data drifts significantly from the baseline (e.g., new customer demographics, changed product ranges), it generates violations and sends CloudWatch alerts. SageMaker Clarify is for pre-deployment bias analysis and explainability — not ongoing production data drift detection. Data Wrangler is a data preparation tool, not a monitoring service. Ground Truth is for human data labeling — not production monitoring.